

AMR HAMDY MOHAMED

IoT & Smart City Solutions Engineer

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PROFESSIONAL SUMMARY

IoT and Smart City engineering leader with over 11 years of experience designing, deploying, and integrating IoT-enabled solutions across smart mobility, smart infrastructure, and smart city domains in Saudi Arabia and Egypt. Skilled in end-to-end solution design, system integration, device management, and cloud platform onboarding, with hands-on expertise across a wide range of communication technologies including NB-IoT, LTE-M, LoRaWAN, GSM/GPRS, BLE, and satellite communication. Proven track record delivering large-scale, multi-stakeholder programs — from national smart capital initiatives to smart port and fleet management platforms — while partnering with cross-functional and non-technical teams to translate complex technical concepts into business outcomes.

CORE COMPETENCIES

- Solution Design & Integration:** End-to-end IoT solution design, system integration, and API integration across multi-vendor platforms
- Device & Platform Management:** Device onboarding, configuration, lifecycle management, and cloud platform administration (Thingsboard, Azure IoT, MasterofThings, Honeywell City Suite)
- Connectivity & Communication:** NB-IoT, LTE-M, LoRaWAN, satellite communication, 3G/4G, GSM/GPRS, BLE, Wi-Fi, CAN, SPI, I2C, UART
- Protocols & Standards:** HTTP, MQTT, MODBUS, and other IoT interoperability standards
- Smart City Domains:** Smart Mobility, Smart Lighting, Smart Parking, Smart Waste Management, Asset Management, Intelligent Operations
- Digital Twin:** Virtual asset modeling, real-time data integration, simulation, and predictive analytics
- Programming & Tools:** C, C++, Python, Node-RED, MATLAB/Simulink, DevOps basics
- Hardware & Embedded:** PCB design (Eagle, Altium, Proteus), high-speed circuit design, lab instrumentation, embedded C for microcontrollers, RTOS

PROFESSIONAL EXPERIENCE

Engineering Lead – IoT Integrated Devices

Mar 2024 – Present

IoT Squared, Riyadh, Saudi Arabia

- Lead end-to-end IoT device integration with cloud platforms, overseeing onboarding, configuration, and full device lifecycle management across multiple platforms.
- Design, deploy, and integrate IoT-enabled solutions supporting city-scale digital transformation across Smart Cities, Smart Mobility, and Smart Solutions applications.
- Partner with cross-functional teams to align solution architecture with platform capabilities and stakeholder requirements.

Smart Cities Expert (IoT Specialist)

May 2022 – Feb 2024

Honeywell, Cairo, Egypt

- Led integration of IoT endpoints with platforms to enable seamless device-to-platform operations.
- Managed CCP, IoT Ops, Asset Management, Smart Lighting, and Smart Parking applications across multiple deployments.
- Recognized with the Certificate of Engineering Excellence – Employee of the Quarter (Q1 2023) for Egypt and North Africa.

Senior IoT Solutions Engineer

May 2018 – Apr 2022

SSTM, Cairo, Egypt

- Delivered Smart City and IoT solutions across device integration, platform configuration, and field deployment.

- Developed and supported Smart Home IoT applications.

KEY PROJECTS

Smart City Program – Administrative Capital for Urban Development (ACUD), New Capital, Cairo

- COC: Built the foundation of the smart capital program, integrating ICT and IoT to digitize and automate city services.
- CCC: Delivered a security monitoring solution for sensitive areas, enhancing crime prevention and response for the new capital.

Smart Waste Management – Riyadh Region Municipality

- Delivered a smart waste management platform and mobile app for real-time bin monitoring, collection scheduling, and operational reporting, enabling data-driven efficiency improvements.

Intelligent Operations – STC

- Designed and deployed IoT sensor solutions for monitoring critical infrastructure (manholes, handholes, telecom cabinets), enabling intrusion detection, predictive maintenance, and remote control.

Container Tracking & Management – ZATCA (Zakat, Tax and Customs Authority)

- Designed and implemented IoT-based container tracking at Jeddah Islamic Port to enhance real-time visibility, security, and logistics efficiency, supporting Saudi Vision 2030.

Fleet Control Services (FCS)

- Delivered fleet control and smart mobility services to 4,000+ customers across Saudi Arabia via telematics, IoT trackers, and data platform integration.

Smart City Solutions Portfolio

- Smart Lighting – ZED Park, Sheikh Zayed City, Egypt
- Smart Waste Management – Borg El Arab Technology Park (with NTRA)
- Smart Parking & Smart Waste Management – Information Technology Institute (ITI)

Additional Notable Projects

- PMIC Evaluation Kit (iPMIC) – Atlazo, USA: designed an intelligent PMIC EVK for hearable, wearable, and ultra-low-power IoT devices.
- NB-IoT Platform & Devices – Vodafone International: designed a platform and NB-IoT devices (3G/4G/NB-IoT) for remote network measurement and configuration.
- Car Access Control System – Zoomcar: remote vehicle control via Bluetooth, 3G, and 4G.
- OpenLock – IoTBlue, Spain: mobile-app-based remote door access control system for offices, garages, and buildings.
- IOTIK – official IoT training kit for Information Technology Institute (ITI) students.
- Smart Shelves – weight/inventory sensing shelves with real-time cloud updates.
- CoolTeck – 3G-connected wireless temperature sensor with SMS/email configuration and emergency alerts.
- Machine learning-based OCR system; wireless line-tracking vehicle; VHDL-based Intel 8085 microprocessor on FPGA.

EDUCATION

BSc, Electronics and Communications Engineering

Cairo University — Graduated May 2015 (Graduation Project Grade: A+)

- Graduation Project: Visible Light Communication System over FPGA — funded by ITIDA and NTRA; awarded 2nd Prize at Inno-vent 2015 (ICECS, sponsored by IEEE, AUC, Zewail City, Mentor Graphics).

CERTIFICATIONS & DIPLOMAS

- Certificate of Smart Cities – Management of Smart Urban Infrastructures, École Polytechnique Fédérale de Lausanne (EPFL)
- Embedded Systems Diploma – C Programming & Data Structures, Embedded Systems Concepts, Embedded C for Microcontrollers, Embedded Systems Software Design (UML), HW/SW Co-Design, RTOS & Scheduling, Software Engineering for Embedded Systems, Verification & Testing, Embedded Systems Development Tools
- Software Testing Concepts and Techniques Diploma – Online ITI Scholarship
- Communications, Presentation, and Interviewing Skills – Systel Training Center (Smart Village), in collaboration with New Horizons

ACHIEVEMENTS & AWARDS

- Certificate of Engineering Excellence – Employee of the Quarter, Q1 2023, Honeywell (Egypt & North Africa)
- 2nd Prize, Inno-vent 2015 Graduation Project Competition – ICECS, sponsored by IEEE, AUC, Zewail City, and Mentor Graphics

ADDITIONAL INFORMATION

Languages: Arabic (Native), English (Professional Working Proficiency)